* 다음 항등에 대한 진리표를 보여라.

a. $F = XYZ + Y' + XZ$

X	Y	Z	F
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

b. $G = XZ + (X'+Y)(X+Y)$

X	Y	Z	G
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

c. $H = XY + WYZ + X'Z$

W	X	Y	Z	H
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	1
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

* 최소 표의항 식으로 줄여라

(a) $xy + xyz + x'$

$= xy(1+z) + x'$

$= xy + x'$

$= y + x'$

(b) $xyz + xy'z$

$= xz$

(c) $x' + y' + xyz'$

$= (x'+y') + xyz'$

$= (xy)' + xyz'$

$= (xy)' + z'$

$= x' + y' + z'$

(d) $abc + a' + a'b$

$= abc + a'(1+b)$

$= abc + a'$

$= a' + bc$

(e) $xyz' + y'z + x$

$= x(1+yz') + y'z$

$= x + y'z$

(f) $xyz + xy'z + xyz' + x'y'z'$

$= xz + xyz' + x'y'z'$

$= x(z+yz') + x'y'z'$

$= x(z+y) + x'y'z'$

$= xz + xy + x'y'z'$

(g) $x+y+z' + xyz' + xyz$

$= x+y+z' + xy$

$= x(1+y) + y + z'$

$= x+y+z'$

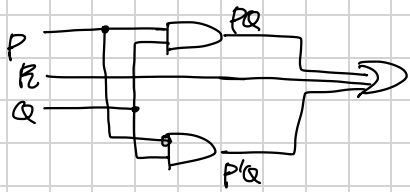
(h) $abc + a'b'c + a'bc + ab'c$

$= ac + a'c$

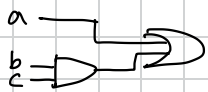
$= c$

* 다음 함수를 구현하기 위한 시스템의 회로도를 AND, OR, NOT 게이트를 사용하여 나타내어라

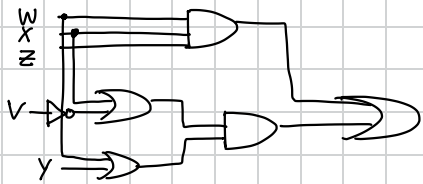
(a) $PQ+R+P'Q$



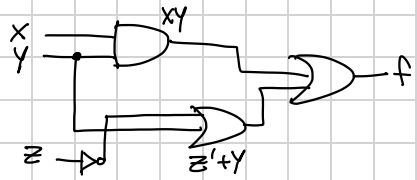
(b) $a+bc$



C. $WXZ+(X+V')(W+Y)$

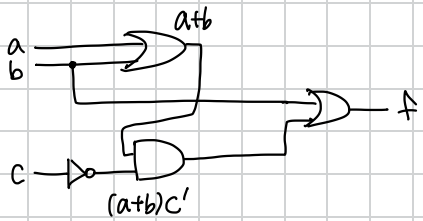


(a)



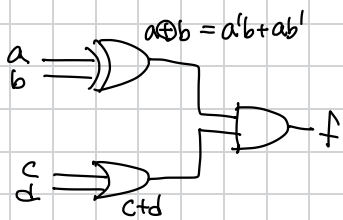
$f = xy + z' + y = y + z'$

(b)



$f = b + (a+b)c' = b + ac' + bc' = b + ac'$

(c)

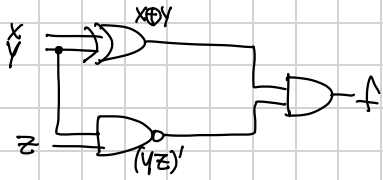


$f = (a \oplus b)(c+d)$

$= (a'b + ab')(c+d)$

$= a'bc + ab'c + a'bd + ab'd$

(d)



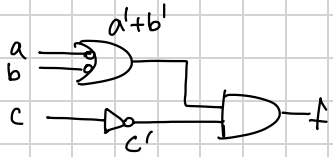
$f = (x \oplus y)(yz)'$

$= (x'y + xy')(y' + z')$

$= xy' + x'yz' + xy'z'$

$= xy' + x'yz'$

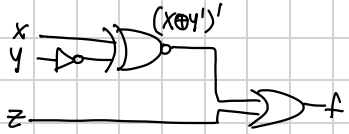
(e)



$f = (a'+b')c'$

$= a'c' + b'c'$

(f)



$f = (x \oplus y')' + z$

$= (x'y' + xy)' + z$

$= (x'y')'(xy)' + z$

$= (x+y)(x'+y') + z$

$= xy' + x'y + z$